

UNIT-5

Electrical Installations

→ LT Switch gears

a) wires and cables

b) Fuses

c) MCB's & MCCB's

d) ELCB.

→ Power factor Improvement
using static Capacitors

→ Energy Consumption calculations

→ Earthing

Introduction to LT Switch gears:

→ switch gear is a component which
~~contains~~ consists of switching &
protecting devices that helps to
protect Electrical Appliances from
abnormal Electrical conditions
such as over voltages short
circuits etc

→ LT stands for "Low tension" i.e.
a voltage range b/w 230 to
400 volts.

→ The LI switch gears components are wires, cables, switch Fuse unit (SFU), MCB's & MCCB's & ELCB.

Wires & Cables:

WIRE: An Electrical wire is defined as single conductor or a group of thin conductor strands covered by insulating material to protect from electrical shocks.

Factors affecting choice of wires:

- 1) Wire size
- 2) Wire lettering
- 3) Colour codes
- 4) Ampacity & Wattage.

⇒ Wire Size:

The standard size of for the difference wires is prescribed by American wire ~~Gauge~~ gauge (AWG) system.

⇒ The diameter of the wire in AWG is converted to mm using the formula.

$$d = 0.127 \times 92$$

where d = diameter of the wire in mm. diameter of wire in the units of (AWG)

2) Wire Lettering:

The Letters written on the outer covering determine the type of the insulation of the material.

- T = Thermo plastic insulation
- TH = High heat Resistance insulation
- H = Heat resistance insulation
- W = wet location insulation
- N = Nylon coating
- X = Flame resistant insulation.

3) Colour Codes:

The different colours available for insulation material are as follows:

1. Red / Brown ⇒ Single phase line
2. Blue / Black ⇒ Single phase neutral.
3. Green ⇒ Single phase earthing
4. Red, Yellow, Blue ⇒ 3 ϕ lines [R, Y, B]

(4) Ampacity & Wattage:
Ampacity and wattage are the two important factors that decide the size of the wire for a particular application. Ampacity implies maximum value of current a conducting wire can transfer and wattage implies the maximum power the conducting wire withstands and transfers.

Types of Wires:

There are two types of wires
1. Solid wire 2. Stranded wire.

→ A solid wire is a single conductor made of copper covered by a protective coloured insulation.

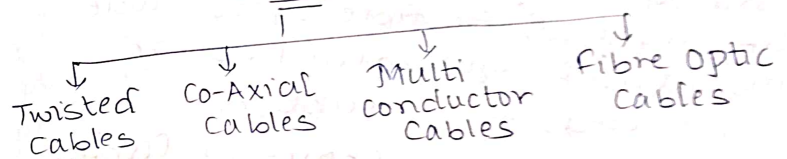

→ A stranded wire contains many thin strands of copper wires twisted and covered with coloured insulation. The stranded wires have the capacity to withstand fluctuations in current and also have long life.

CABLES:

When two or more wires are bounded together, twisted together, braided together and sheathed with protective insulation forms cables. Cables are mostly used in power transmission and telecommunication applications.

Types of Cables:

CABLES:



⇒ Twisted Cables:

When two cables are twisted to carry signals from one point to another forms a twisted cable.
Ex: transmission lines.

⇒ Co-Axial Cables has an inner solid conductor surrounded by a protective insulation sheath.
Ex: TV cables.

⇒ Multiconductor Cables are those cables which consists of two or more conductors insulated from each other. They are widely used in computers, communication systems, sound and picture transmission.

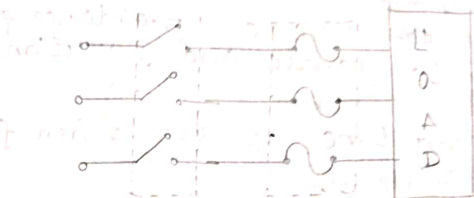
⇒ Fibre Optic Cables are internet cables having high band width. Optical fibre refers to a technology that transmits information as light pulses along glass or plastic fibre. Optical fibre cables are generally used for long distance communication. The material for fibre optic cable insulation is generally PVC or vulcanized, tough rubber.

Factor Affecting choice of Cable:

The choice of cables depends on the following

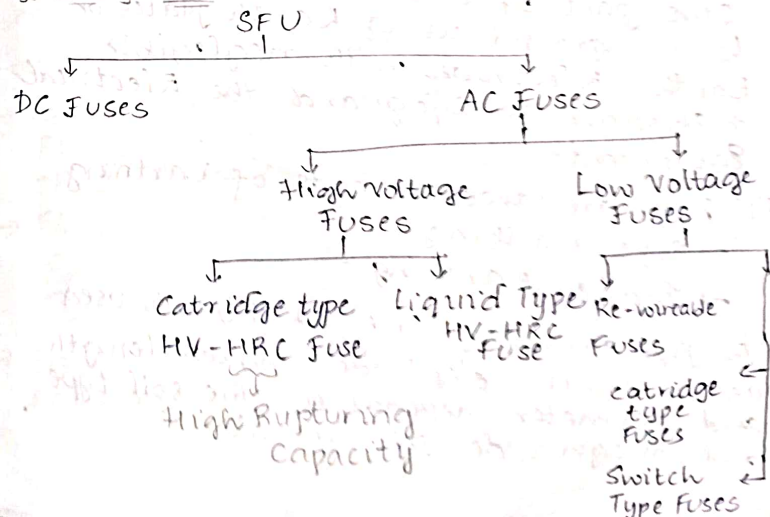
1. Type of Application
2. Duration of cable life.
3. Costs [maintenance cost included]
4. The cable should be resistive to harsh weather & chemicals.
5. The cable should be easily accessible [for Troubleshooting]

SWITCH FUSE UNIT [SFU]



1. The SFU is a low voltage A.C. Fuse that is used to protect electrical devices from overload or shortcircuit conditions.
2. The domestic fuses have a rating b/n 30Amps to 800Amps where as industrial fuses have a rating upto 46 kAmps.
3. The SFU has the capability of withstanding till the fault current reaches 3 times its rated value beyond which the SFU safely breaks circuit protecting the building or electrical installations.

Types of SFU:



Advantages of SFU:

1. It is cause effective and maintenance free.
2. It is easy to operate and pollution free.
3. Its operating time is 2sec when fault occurs.
4. Rewiring & replacing in case of damage makes the SFU maintenance free.
5. When the fuses are connected in Series they have to be labelled for identification.

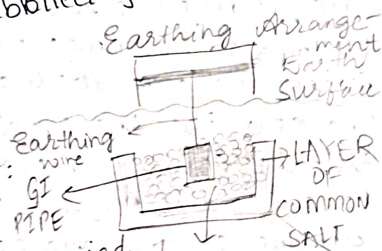
★ EARTHING:

(i) Pipe Earthing:

Earthing refers to connecting the electrically conductive part of electrical equipment to the ground using Earth plates or Earth Electrodes with negligible resistance to safeguard the Electrical Equipment.

- (ii) There are two methods of Earthing
- Pipe Earthing.
 - Plate Earthing.

In Pipe earthing The GI pipe is used as an Earth electrode whose length and diameter depends on the soil type and magnitude of current.



The standard diameter of the pipe should be greater than 3.8 cm and length of the pipe should be 2 meters for ordinary soil, 2.75 m for dry soil & 1.75 m for rocky soil. The GI pipe is vertically placed and buried to a depth of nearly 5mts ^(depending on soil types). Charcoal & layer of common salt is used to surround the G.I pipe which helps in protecting from the moisture present in the Earth, also they help in decreasing the Earth's resistance.

(vi) In Plate Earthing a G.I plate or copper plate is used instead of pipe. The dimensions of the G.I pipe is $60\text{ cm} \times 60\text{ cm} \times 6.35\text{ mm}$ [$l \times b \times h$] for copper plate $60\text{ cm} \times 60\text{ cm} \times 3.2\text{ mm}$ [$l \times b \times h$]

These plates are covered upto 15cm with charcoal & common salt.

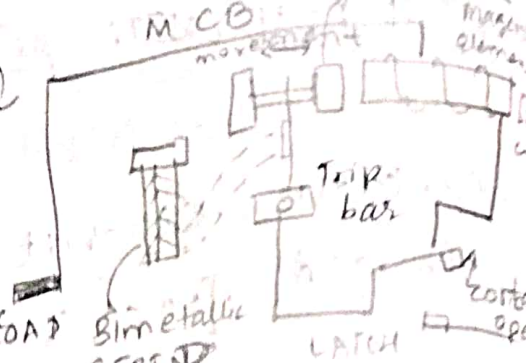
Factors to be Remembered for Earthing:

- Distance :- The distance b/w the Earth Electrode & Electrical Installation should be greater than 1.5m.
- Electrodes :- The material used for Earth Electrode or plates should be Galvanised Iron or Copper.
- Earth Resistance :- Ensure the Earth resistance to be 5-2.

Need for Earthing:

1. Earthing protects electrical equipment & buildings from over voltages occurring due to lightening.
2. Earthing prevents fire accidents due to short circuits.
3. Earthing protects human lives by protecting Electrical Installation from fault currents. By keeping and maintaining a const. Voltage when fault occurs.

Miniature Circuit Board [MCB]



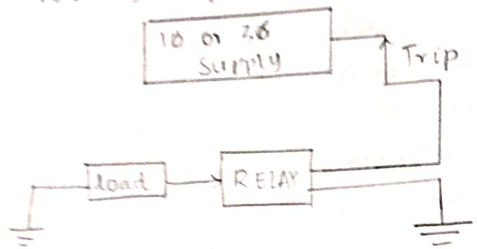
- 1) The miniature circuit Board is an electromagnetic device which protects the electrical board from over currents i.e. short circuit or overload conditions.
- 2) The MCB uses Thermal & Magnetic Trip units to open the contact during any fault condition.
- 3) Magnetic Tripping occurs when a short circuit condition exists in electrical circuit. The S.C current energises the magnetic element creating a magnetic field, attracting the Plunger in such a way that the force exerted on the Trip bar causes the contact to open.
- 4) Thermal tripping occurs when overload condition occurs in the electrical circuit causing a large fault current to flow through the circuit. This fault current generates heat in the bimetallic strip, resulting in thermal expansion of the bimetallic strip which exerts the force on the Trip bar causing the contact to open.

Advantages of MCB: (2m)

- 1) It does not require any replacement of parts.
- 2) When overcurrent is detected in the circuit.
- 3) MCB offers operational Safety and also easier operation when compare to conventional fuses.
- 4) MCB's are rated to withstand overload currents upto 5 to 15 times of rated current.
- 5) In domestic applications the rated current for MCBs is 10 Amperes (upto 100A).
- 6) The Operating Time of MCB to trip the main supply for S.C fault is less than 0.5 msecs & for over loads it's usually b/w 2secs to 2mins.

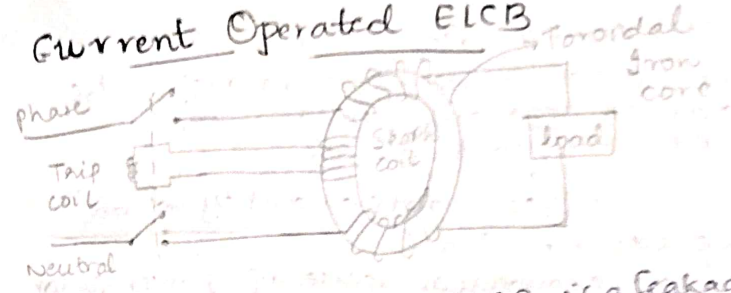
EARTH LEAKAGE CKT BREAKER [ELCB]

Voltage Operated ELCB



The ELCB is a safety device which detects the leakage current flowing to the Earth directly & Trips the power supply it prevents Electric shocks. The principle of operation of voltage operated ELCB is based on voltage difference to detect Earth Leakage. It consists of relay coil whose one terminal is connected to the Load & the other Terminal is connected to the Earth and the as shown in fig if any fault current occurs then a voltage diff. appears across the terminals of relay which results in tripping of the main supply.

Current Operated ELCB



In current operated ELCB if a leakage current is detected in the load the current flowing through the phase & neutral differs which results in residual current which gives rise to residual emf in the search coil, due to this induced residual emf the trip coil gets energized which isolates the Load from the main power supply. due to this residual emf which results in tripping of the main supply another name for current operated ELCB is "RCCB". i.e. "Residual Current Circuit Breaker".

Power Factor Improvement

Power factor is defined as ratio of active power to apparent power.

$$P.F = \frac{VI \cos \phi}{VI} = \cos \phi$$

It is a unitless quantity whose value lies b/w 0 & 1.

The most economical value of power factor is b/w 0.9 to 0.95

Relationship b/w Power Factor & Current:

$$\left. \begin{array}{l} \text{Active} \\ \text{or} \\ \text{Real Power} \end{array} \right\} = P = VI \cos \phi$$

$$\cos \phi = P.F. \text{ (Power factor)}$$

$$\cos \phi = \frac{P}{VI} \text{ Since } P, V = \text{constants}$$

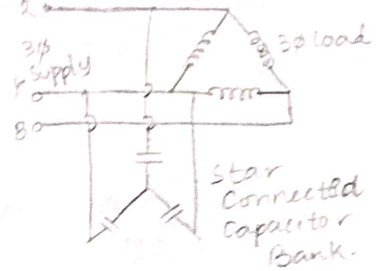
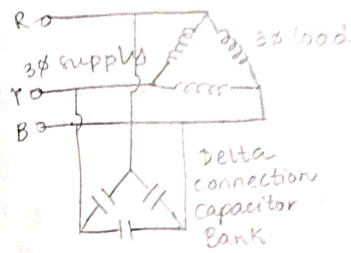
$$\cos \phi \propto \frac{1}{I}$$

$$\left. \begin{array}{l} \cos \phi \downarrow \Rightarrow I \uparrow \\ \cos \phi \uparrow \Rightarrow I \downarrow \end{array} \right\}$$

Causes of Low Power factor:

1. Single Phase & 3-Phase Induction motors are purely inductive in nature where the current lags the voltage by 90° .
2. Transformers, Lamps, furnaces are also other inductive equipments which cause low Power factor.
3. Improper wiring, usage of low grade wires, cables also cause imbalance in the PF.
4. Electromagnetic interferences, harmonics, also reduced Power factor.

(a) Power factor Improvement using Static Capacitor.

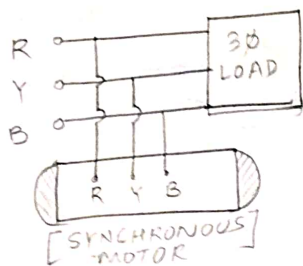


- 1) If the power factor of the system is low, then it can be adjusted by connecting static capacitor bank.
- 2) For 3-phase supply the capacitor bank can be connected either in star or Delta connection as shown in fig.
- 3) The use of capacitors improves the power factor because they exhibit leading PF i.e. current leads the voltage.

Advantages of Using Static Capacitor:

- Ease of installation and low on maintenance for both single & 3-phase load.
- Less weight and hence the system will be less bulky.
- In small distribution substations capacitance are connected in star or delta and at the supply lines with the range 500 - 3000 KVAR.
- The main disadvantage of using Static Capacitor is the life span. Usually the capacitor bank has to be replaced every 5 yrs.

Power Factor Improvement using SYNCHRONOUS CONDENSER



When a synchronous motor operates at a no-load & with over excitation then it is called as synchronous condenser. This over excited synchronous motor when connected across the supply lines draws leading current there by improving the power factor. This method is generally used to improve the power factor in large scale industries which employ high machinery.

ADVANTAGES:

- 1) Life span of synchronous condenser is more compare to capacitor.
- 2) In 3-phase system replacement of synchronous condenser is easier when compare to star or Delta connected capacitor bank.
- 3) Electromagnetic Interferences, harmonics do not have any effect on the synchronous condenser.

DISADVANTAGES:

- 1) High maintenance cost.
- 2) Noise pollution.
- 3) Unsuitable for chemical industries.

ENERGY CONSUMPTION CALCULATION

Energy and power are closely related quantities.

$$\text{Power} = \frac{\text{Energy}}{\text{Time}}$$

$$\Rightarrow \text{Energy} = \text{Power} \times \text{Time}$$

Units are (KW-hr)

$$\boxed{1 \text{ unit of Energy} = 1 \text{ KW-hr}}$$

- (1) A consumer uses 10KW geyser, 6KW Furnace, 500W bulb for 15 hrs daily. Calculate the total energy for one day and one month of 30 days?

$$\begin{aligned} \text{Geysor} &= 10 \times 15 = 150 \text{ KW-hr} \\ \text{Furnace} &= 6 \times 15 = 90 \text{ KW-hr} \\ \text{Bulb} &= 500 \times 0.5 \times 15 = 37.5 \text{ KW-hr} \end{aligned}$$

$$[500 \text{ W} = 0.5 \text{ KW}]$$

$$1 \text{ DAY} = 247.5 \text{ KW-hr}$$

$$30 \text{ DAYS} = 247.5 \times 30 = 7425 \text{ KW-hr}$$

2) In a house 5 Tubelights of each 40 watts,
 3 Fans of each 150W & Refrigerator
 of 200W are used for 20hrs per day.
 What will be the power bill for the
 month of October. if the chargers
 are 1 unit = Rs 5/-

Q2 OCTOBER = 31 Days.

$$\text{load-1} = \text{Tubelights} = 5 \times 40 = 200W = 0.2 \times 20 = 4 \text{ KW-h}$$

$$\text{load-2} = \text{Fans} = 3 \times 150 = 450W = 0.45 \text{ KW} \\ = 0.45 \times 20 = 9 \text{ KW-h}$$

$$\text{load-3} = \text{Refrigerator} = 1 \times 200W = 0.2 \text{ KW} \\ = 0.2 \times 20 = 4 \text{ KW-h}$$

$$\text{Total Energy per day} = 17 \text{ KW-h}$$

$$\text{Total Energy for 31 days} = 17 \times 31 \\ = 527 \text{ KW-h}$$

$$1 \text{ KW-h is 1 unit} = 5$$

$$\text{Total Power Bill} = 527 \times 5 = 2635 \text{ /-} \\ = \text{Rs} - 2635 \text{ /-}$$